

D. The LCM Weave

time limit per test: 1 second
memory limit per test: 256 megabytes

Let there be a function defined as: $f(X, Y) = \left\lfloor \frac{\text{lcm}(X, Y)}{X \cdot Y} \right\rfloor$

You are given an array A of N integers. Your task is to compute the sum of $f(X, Y)$ over all possible unordered pairs (X, Y) formed from the array.

† An unordered pairs refers to a selection of two distinct indices i, j such that $i \neq j$, and the corresponding elements are $X = A[i]$, $Y = A[j]$. Note that the values X, Y may be equal, **but the indices must be different**. Each unique set of indices should be considered only once, regardless of the order.

Input

The first line contains a single integer $N(2 \leq N \leq 2 \cdot 10^5)$, the number of elements.

The second line contains N space-separated integers $A_1, A_2, \dots, A_N (1 \leq A_i \leq 10^6)$.

Output

Print the total sum of $f(X, Y)$ over all valid unordered triples.

Example

input	Copy
6 15 4 8 3 2 6	
output	Copy
6	

Replay of BU Intra Department Programming Contest 2025

Finished

Practice



→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
338661854	Sep/15/2025 05:28	Happy New Year!

→ Contest materials

Tutorial (en)